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# What is Geo Data Science?

Édipo Cremon  
Rennan Marujo



# Discipline Catalog

| Código    | Título                        | Créditos |
|-----------|-------------------------------|----------|
| CAP-423-4 | Ciência de Dados Geoespaciais | 4        |

## Trilha **Ciência de Dados Geoespaciais**

Oferecida no 3º Período

Docente(s) Responsável(is): Édipo Henrique Cremon ✉ Rennan de Freitas Bezerra Marujo ✉

## Ementa

Essa disciplina foca em pesquisas e aplicações com dados geospaciais, as quais incluem técnicas de manipulação de imagens, análise de séries temporais e métodos de machine learning para identificar padrões e extrair informações de dados espaço-temporais e de imagens de sensoriamento remoto. De caráter prático, serão abordadas aplicações em que as técnicas de pré-processamento, medidas de similaridade, filtros, extração de atributos, segmentação, clustering, classificação e validação, são empregadas na produção de mapas de uso e cobertura da Terra. Também serão estudados diferentes métodos de análise de séries temporais de imagens de sensoriamento remoto, como TWDTW, TimeSat e BFast. A disciplina é baseada em aulas teóricas e práticas, além de discussão e reprodução de artigos bem conhecidos da literatura, que apresentem o estado da arte nos temas relacionados. Os tópicos principais da disciplina são:

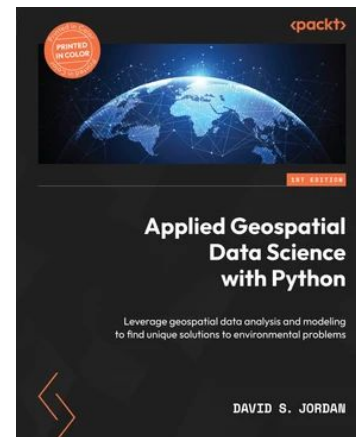
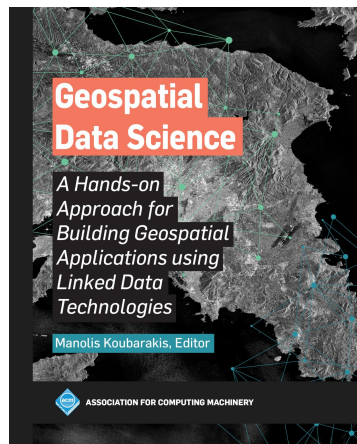
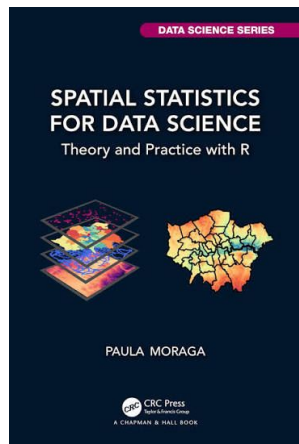
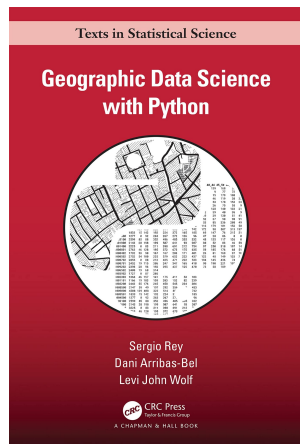
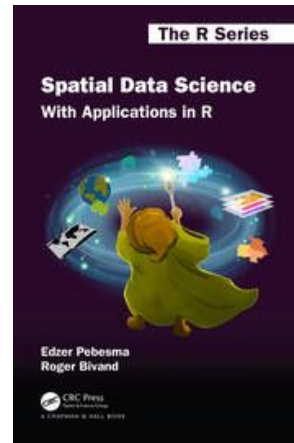
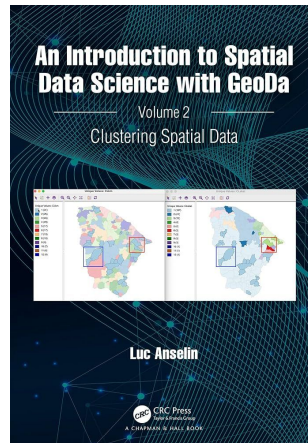
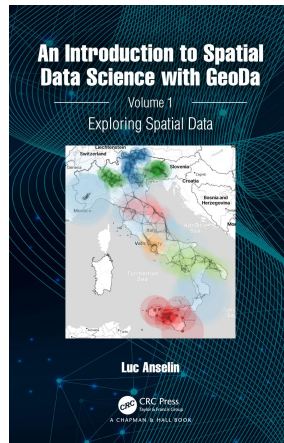
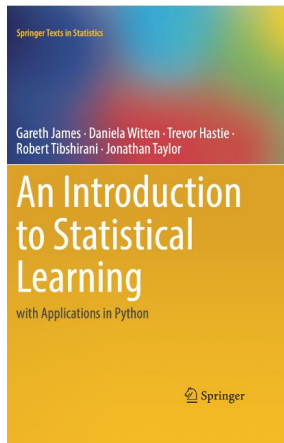
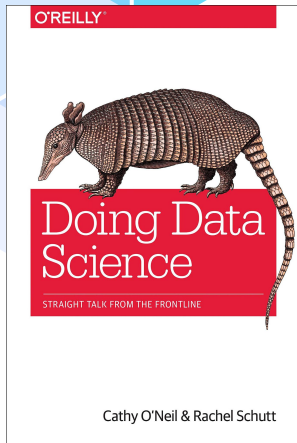
- Introdução a dados espaço-temporais e séries temporais de imagens de sensoriamento remoto;
- Análise de séries temporais de imagens de sensoriamento remoto, incluindo uso de filtros, extração de atributos, medidas de similaridade e clustering;
- Segmentação de Imagens;
- Análise de amostras de uso e cobertura da Terra utilizando redes neurais;
- Classificação de uso e cobertura da Terra utilizando métodos de Machine Learning e Deep Learning;
- Análise de trajetórias de uso e cobertura da Terra;



Édipo Cremon



Rennan Marujo



...and research articles

# Module Structure

## Module 1. Foundations and Methods in Geospatial Data Science

- Data Science vs. Geospatial Data Science
- Coordinate Reference Systems
- Geospatial data formats and Python tools
- Exploratory Spatial Data Analysis
- Geospatial Machine Learning
- Spatial validation and geographically weighted regression
- **WORCAP**

## Module 2. Satellite Image Time Series Analysis

- Introduction to **Satellite Image Time Series (SITS)**
- Spatiotemporal organization of Earth observation data
- Time-series extraction and analysis
- Temporal patterns in remote sensing data
- SITS-based classification and monitoring workflows

## Module 3. Special Topics in Geospatial Data Science

- Spatial clustering and unsupervised learning
- Geospatial segmentation
- Static and interactive geospatial visualization
- Advanced time-series methods: **TWDTW, TIMESAT, WCPMS, BFAST**
- Workshops, seminars, and applied case studies
- **GeoPantanal and GeoInfo activities**

# Course Assessment

Assignments: 20%

Test: 30%

Seminars: 20%

Problem sets: 30%

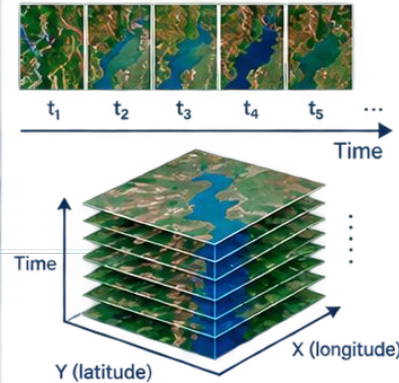
# Why Geo Data Science now?



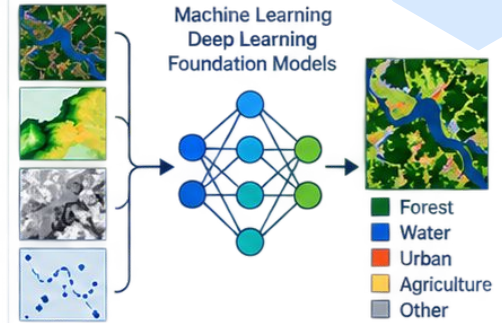
- Earth Observation at unprecedented scale



- Open geospatial data



- Spatiotemporal data and satellite image time series



- New computational and machine learning methods

More geospatial data - but are standard data science methods enough?

# Which element makes a data science problem geospatial?

- A. Large datasets
- B. Geographic coordinates
- C. Spatial relationships relevant to the problem
- D. Producing a map





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# **Quick Recap: What is Data Science?**

# What is Data Science?

Interdisciplinary field combining statistics, computing, and domain knowledge to extract insights and predictions from data.

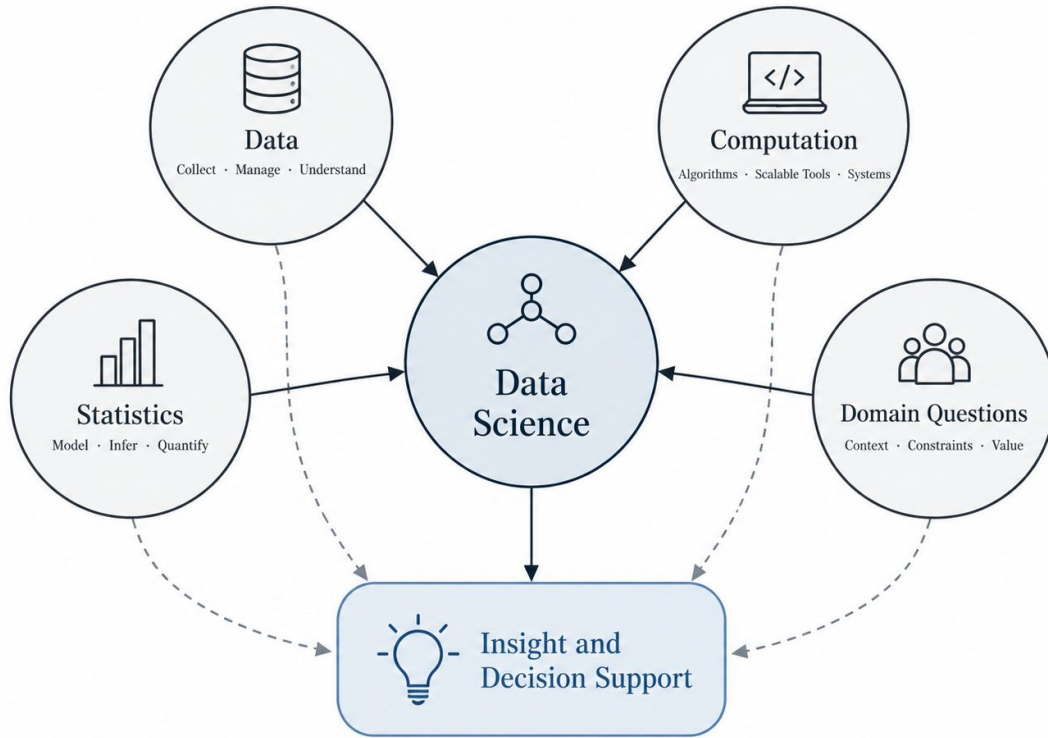


# What is Data Science?

Interdisciplinary field combining statistics, computing, and domain knowledge to extract insights and predictions from data.

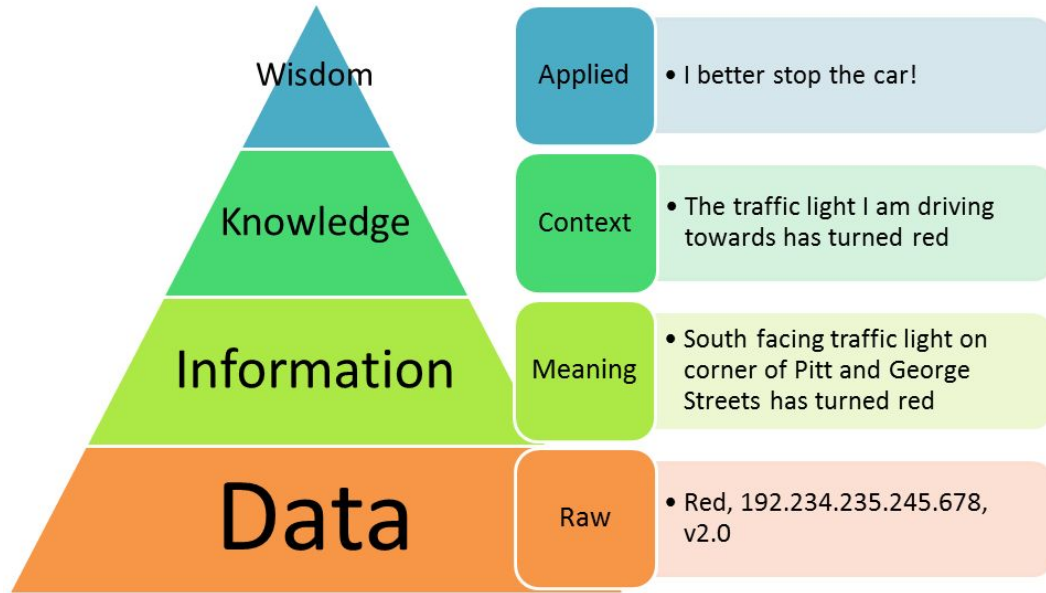


# What makes Data Science distinct?



- Data Science is more than statistics or programming.
- It integrates data, computation, and problem solving.
- It links methods to real analytical questions.
- Its goal is actionable understanding from data.

# Data, Information, and Knowledge

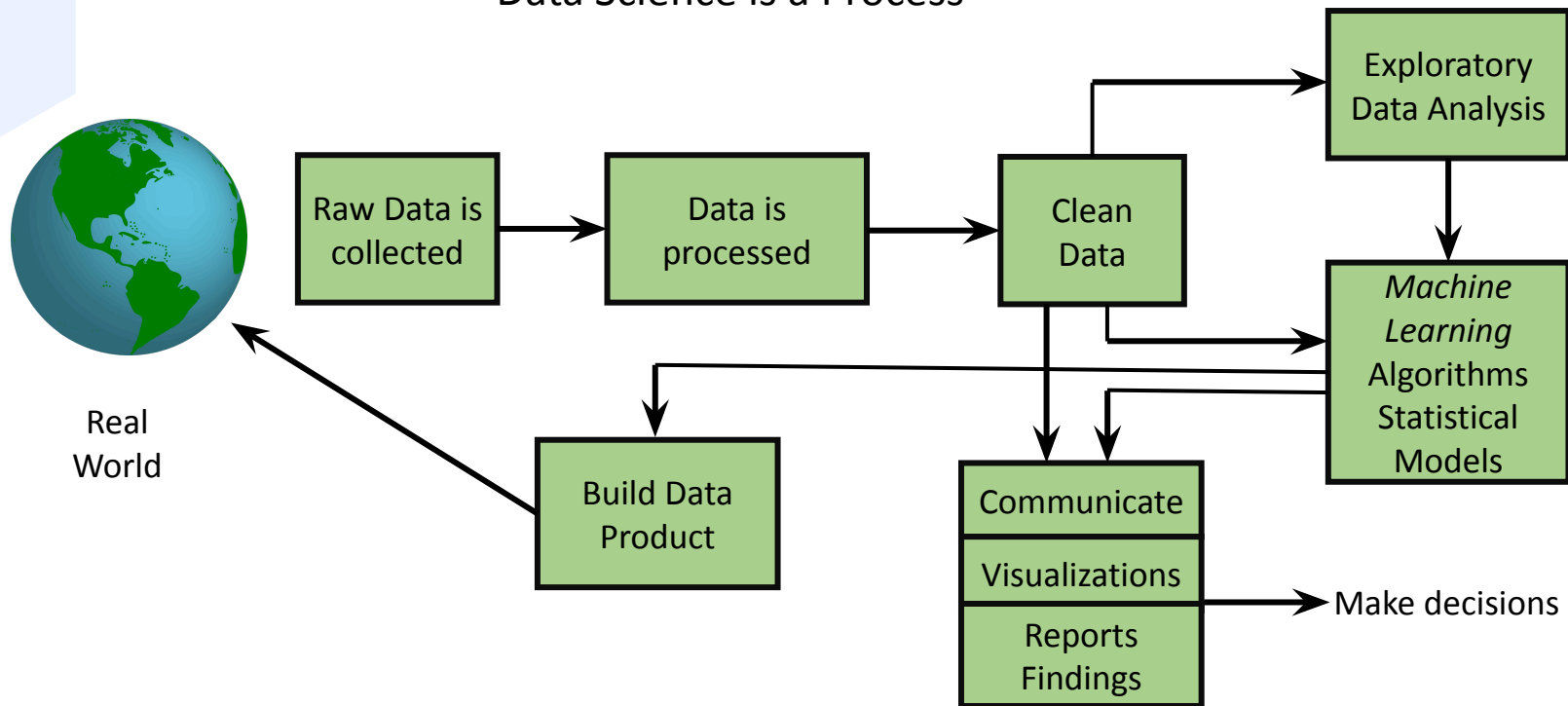


© 2011 Angus McDonald

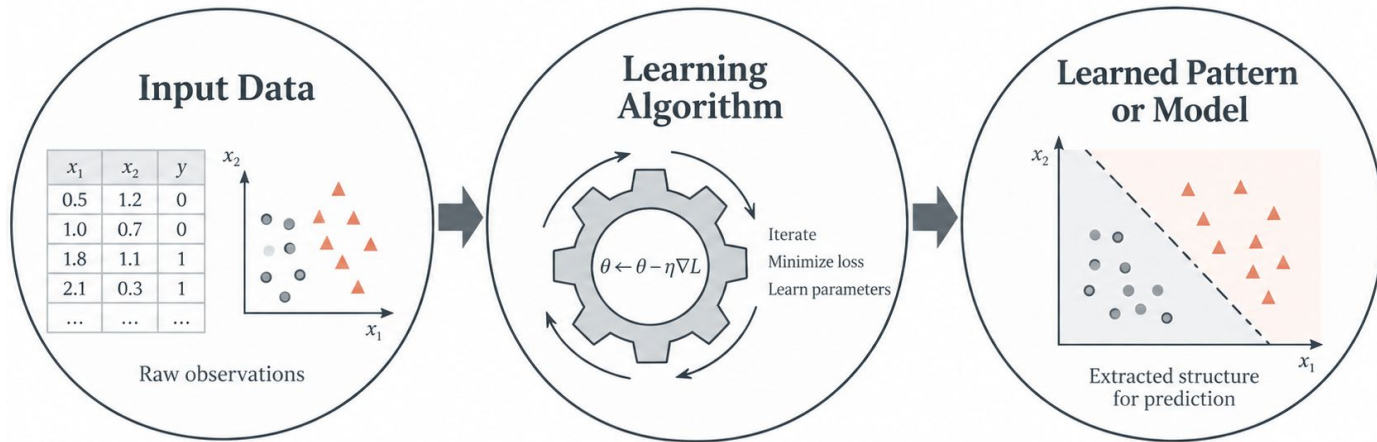
Source: <https://allthingy.com/data-information-knowledge-wisdom/>

# What is Data Science?

Data Science is a Process



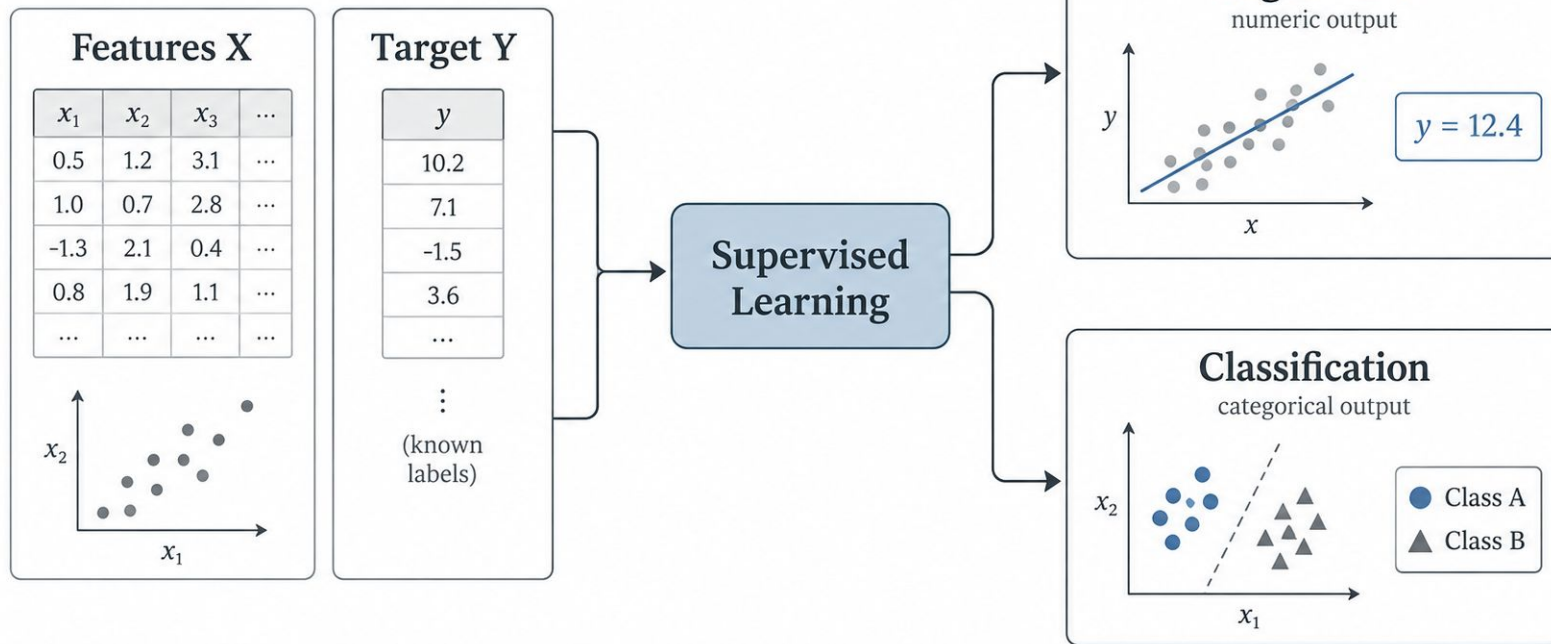
# What does it mean to learn?



- Learning means extracting structure from data.
- Algorithms estimate patterns, rules, or responses.
- Learning depends on data, targets, and assumptions.
- Different tasks require different learning strategies.

# Supervised Learning

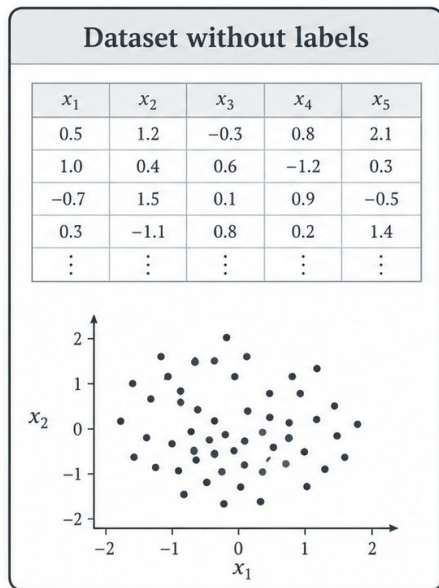
- Supervised learning uses labeled examples.
- Predictors are mapped to a known response.



- Common tasks include regression and classification.

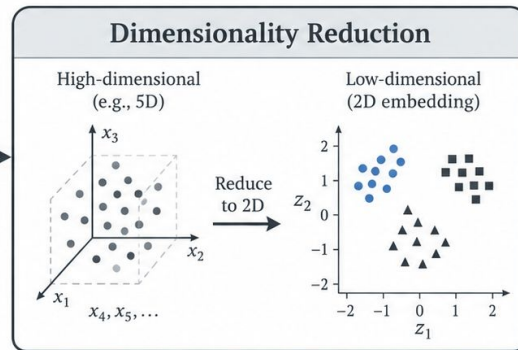
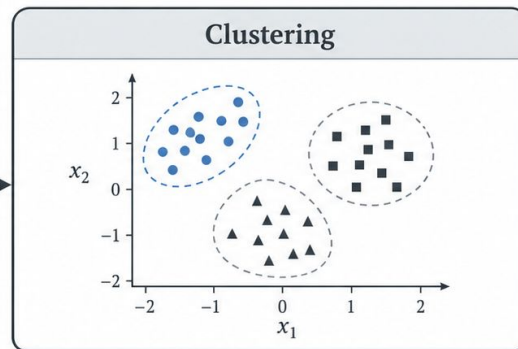
# Unsupervised Learning

- Unsupervised learning has no target variable.
- The goal is to detect structure in observations.



Unsupervised Learning

- It supports clustering, grouping, and dimensionality reduction.





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# QUIZ

# What kind of learning problem is this?

You have labeled satellite pixels representing forest, pasture, and cropland. What type of problem is this?

- A. Regression
- B. Classification
- C. Clustering
- D. Dimensionality reduction

# What kind of learning problem is this?

You have thousands of unlabeled spectral signatures and want to discover groups.

- A. Regression
- B. Classification
- C. Clustering
- D. Spatial interpolation

# What kind of learning problem is this?

You want to predict land surface temperature as a continuous value.

- A. Regression
- B. Classification
- C. Clustering
- D. PCA (Principal Components Analysis)



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**What changes  
when data  
have  
location?**

# Location changes the observation

- Geospatial observations combine attributes and location.
- Location adds context beyond tabular records.

coordinates as ordinary variables?

tabular view

| id | name   | $x_1$ | $x_2$ | ... | $x_n$ | $x$  | $y$  |
|----|--------|-------|-------|-----|-------|------|------|
| 1  | name 1 | 0.42  | 1.21  | ... | 0.18  | 10.5 | 43.1 |
| 2  | name 2 | 0.05  | 0.98  | ... | 0.61  | 10.5 | 43.1 |
| 3  | name 3 | 0.77  | 0.12  | ... | 0.90  | 13.8 | 43.7 |
| 4  | name 4 | 0.26  | 0.83  | ... | 0.37  | 15.7 | 41.3 |

treated as isolated attributes

- Observations can interact through **spatial relations**;
- **Space** becomes **part of the analytical problem**.

# Location changes the observation

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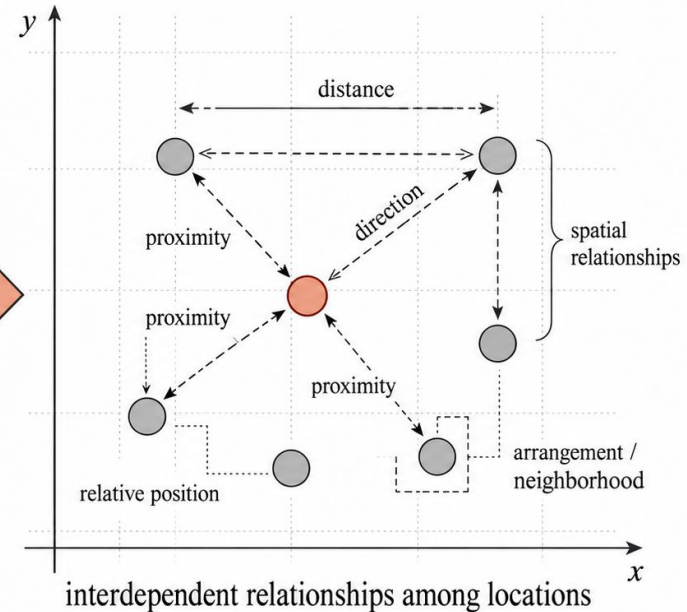
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location reveals  
spatial context

spatial meaning comes from relationships!



treated as isolated attributes

- Observations can interact through **spatial relations**;
- **Space** becomes **part of the analytical problem**.

# What is special about spatial data?

## WHAT IS SPECIAL ABOUT SPATIAL DATA?

### ALTERNATIVE PERSPECTIVES ON SPATIAL DATA ANALYSIS

by

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## What Problem? Spatial Autocorrelation and Geographic Information Science

Michael F. Goodchild

Department of Geography, Center for Spatial Studies, University of California, Santa Barbara, CA

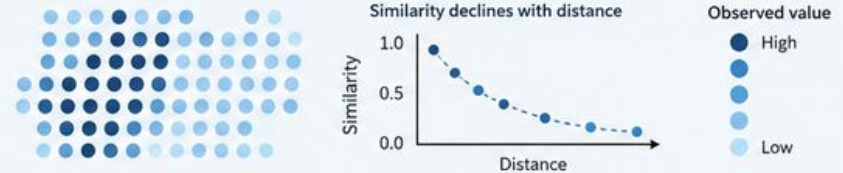
# What is special about spatial data?

## 1 Spatial dependence

Nearby observations may not be independent.

### 1 Spatial dependence

Nearby locations tend to have similar values.

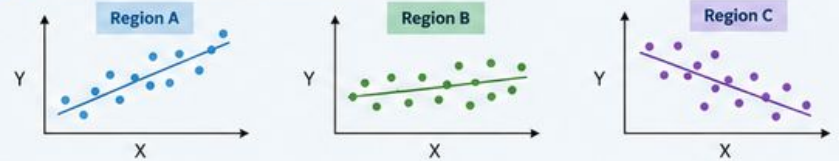


## 2 Spatial heterogeneity

Relationships may across space.

### 2 Spatial heterogeneity

The same relationship can differ across space.



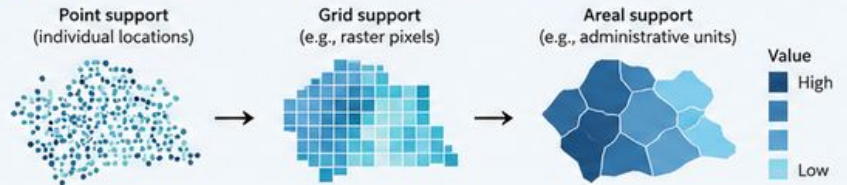
## 3 Scale and support

What we observe depends on the spatial unit or footprint.

Support = spatial unit or footprint over which a variable is observed, measured, or aggregated.

### 3 Scale and support

The same phenomenon can look different at different spatial units.





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# **(Geo)Spatial Data Science and Spatially Explicit Modeling**

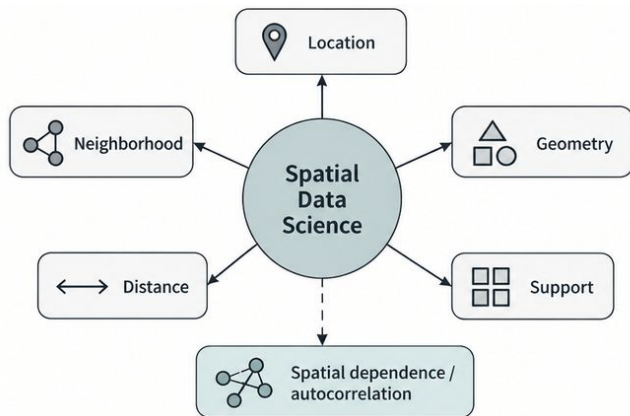
# Spatial, Geospatial, and Geographic Data Science

**An interdisciplinary body of practices for representing, exploring, analyzing, and modeling spatially referenced data—and, when relevant, spatiotemporal data—by combining statistical, computational, and machine learning methods with concepts such as location, distance, neighborhood, geometry, scale, context, and spatial relationships.**

**These three terms denote highly overlapping fields with no rigid boundaries.**



# Spatial Data Science



Applies to many spatial domains  
(e.g. not only Earth)



Earth systems



Urban and  
infrastructure systems



Biological systems

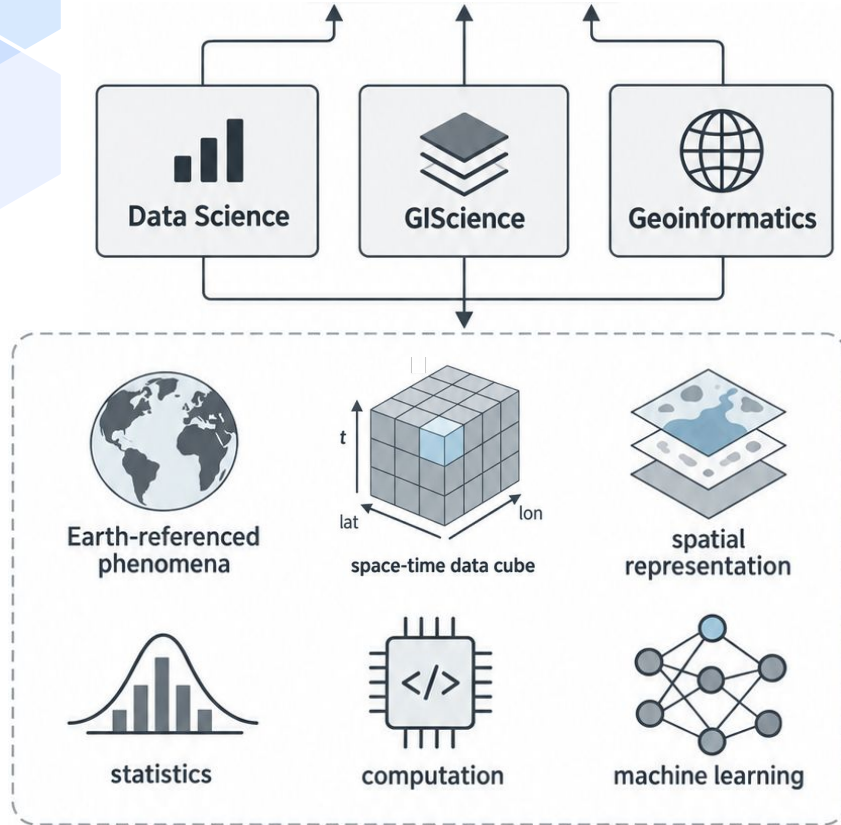


Materials, cells,  
and other spatial domains

- Studies **spatially structured data and processes** in a general sense.
- Focuses on **location, geometry, support, distance, and neighborhood**.
- Treats **spatial dependence and autocorrelation** as part of data structure.
- Applies to spatial domains **beyond strictly Earth-referenced phenomena**.

**References:** [Pebesma and Bivand \(2023\)](#); [Anselin \(2024\)](#); Moraga (2024).

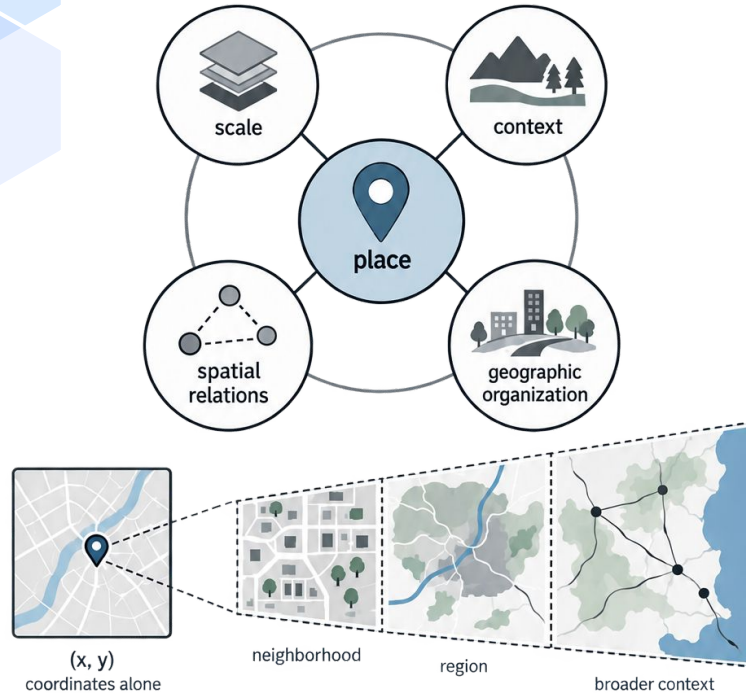
# Geospatial Data Science



- Integrates **Data Science and GIScience**, supported by Geoinformatics.
- Focuses on **Earth-referenced data and phenomena**.
- Frequently incorporates the **temporal dimension** into spatial analysis.
- Combines spatial representation, statistics, computation, and **machine learning**.
- Requires appropriate **coordinate systems, projections, and spatial data structures**.
- May organize large Earth observation collections as **space-time data cubes**.

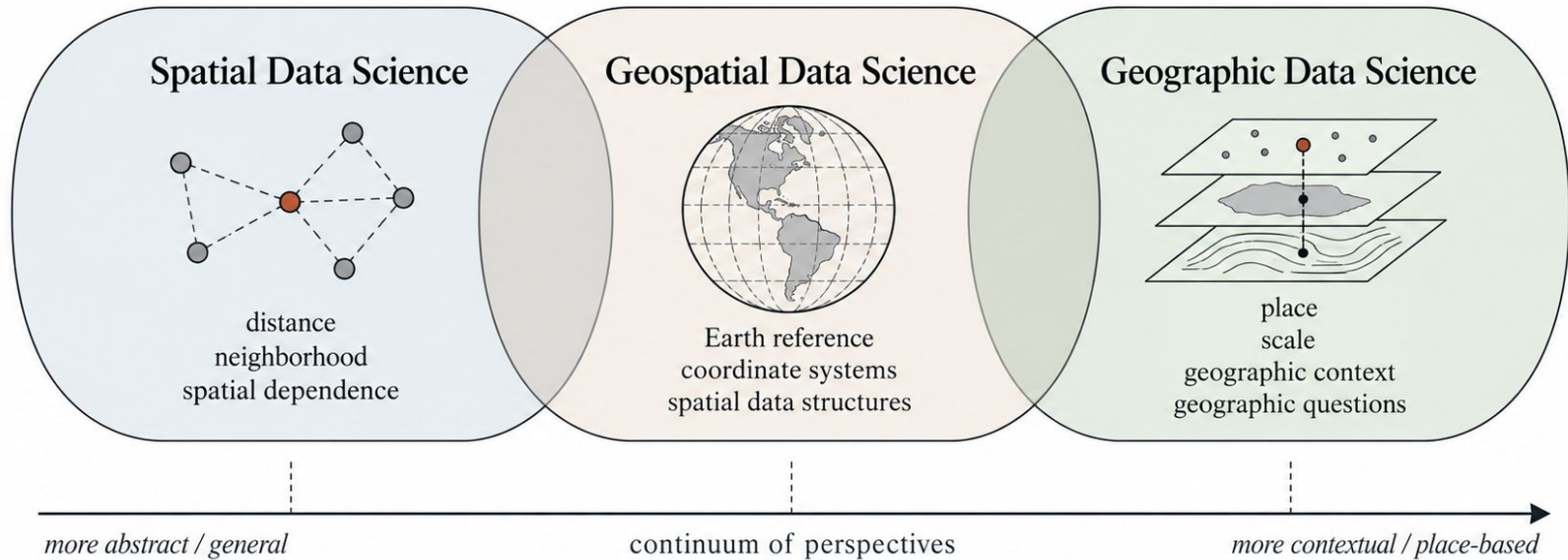
**References:** Eftelioglu et al. (2018); [Percivall \(2020\)](#); [Jordan \(2023\)](#); Koubarakis (2023).

# Geographic Data Science



- Centers the **geographic nature of the research question**.
- Combines **computational reasoning and geographic reasoning**.
- Interprets location through **spatial relations, context, and geographic organization**.
- Emphasizes **place, scale, and spatial configuration**.
- Reframes statistical patterns through a **geographic perspective**.
- Uses concepts such as **location, field, object, event, and network**.

# Spatial, Geospatial, and Geographic Data Science



A heuristic continuum of emphasis - not a taxonomy



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# QUIZ

# Which perspective is emphasized?

Researchers analyze spatial dependence among cells in biological tissue.

- A. Spatial Data Science
- B. Geospatial Data Science
- C. Geographic Data Science

# Which perspective is emphasized?

Researchers process Sentinel-2 imagery using CRS transformations and space-time data cubes.

- A. Spatial Data Science
- B. Geospatial Data Science
- C. Geographic Data Science

# Which perspective is emphasized?

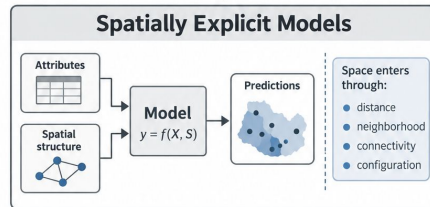
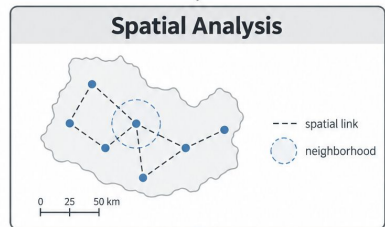
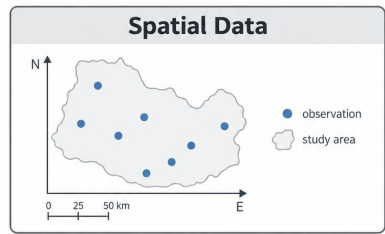
Researchers ask how neighborhood context and scale affect urban segregation patterns.

- A. Spatial Data Science
- B. Geospatial Data Science
- C. Geographic Data Science

# From spatial data to explicit models

- **Spatial data** provide an explicit location reference;
- **Spatial analysis** examines location and relationships among observations;
- **Spatially explicit models** embed space within the model structure;
- Space may enter through **distance, neighborhood, connectivity, or configuration**;
- Spatial explicitness is a **modeling strategy**, not a separate field.

**Space must participate in the model, not merely describe the data.**



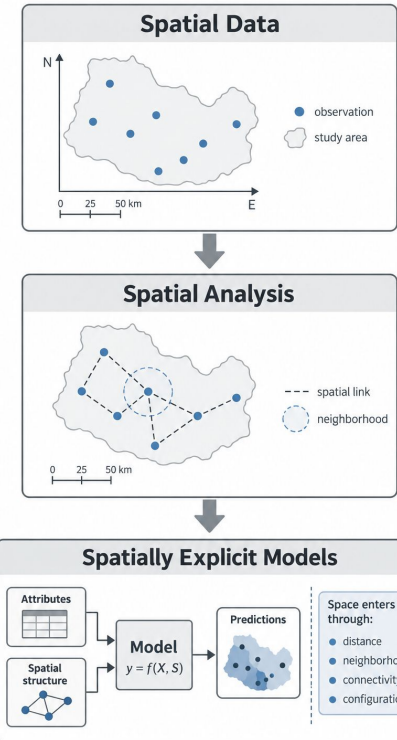
*Spatial explicitness is a modeling strategy, not a separate field.*

**References:** Anselin (1989); Goodchild (2009); Rowe and Arribas-Bel (2026).

# What makes a model spatially explicit?

- Georeferenced data alone do **not** make a model spatially explicit;
- Mapped predictions do **not** demonstrate spatial explicitness;
- Spatial concepts must affect the model's **representation, formulation, or behavior**.
- Results may depend on **where observations are located relative to others**;
- Spatial structure can address **dependence and heterogeneity**.

**A map is an output format, not evidence of a spatially explicit model.**



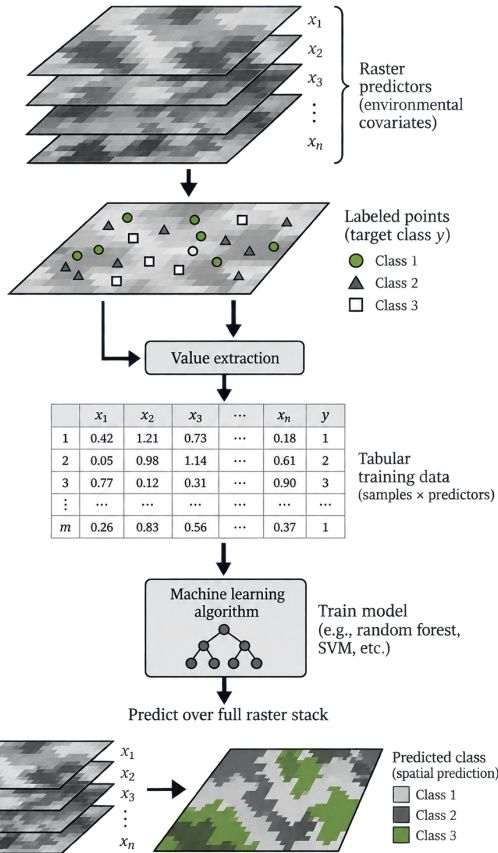
*Spatial explicitness is a modeling strategy, not a separate field.*

**References:** Janowicz et al. (2020); Anselin (1989); Goodchild (2009).

# Is this Geospatial Data Science?

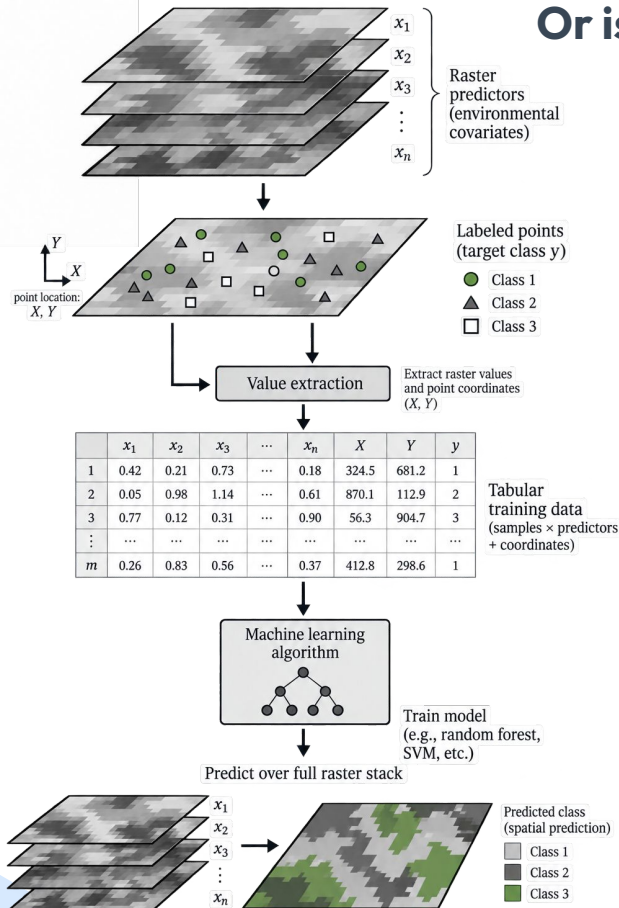
Or is it Data Science applied to spatial prediction?

**Spatial inputs → Tabular learning → Spatial output**



# Is this Geospatial Data Science?

Or is it Data Science applied to spatial prediction?



**Now coordinates ( $X$  and  $Y$ ) are predictors. Does that make it spatially explicit?**

# Which element makes a data science problem geospatial?

- A. Large datasets
- B. Geographic coordinates
- C. Spatial relationships relevant to the problem
- D. Producing a map



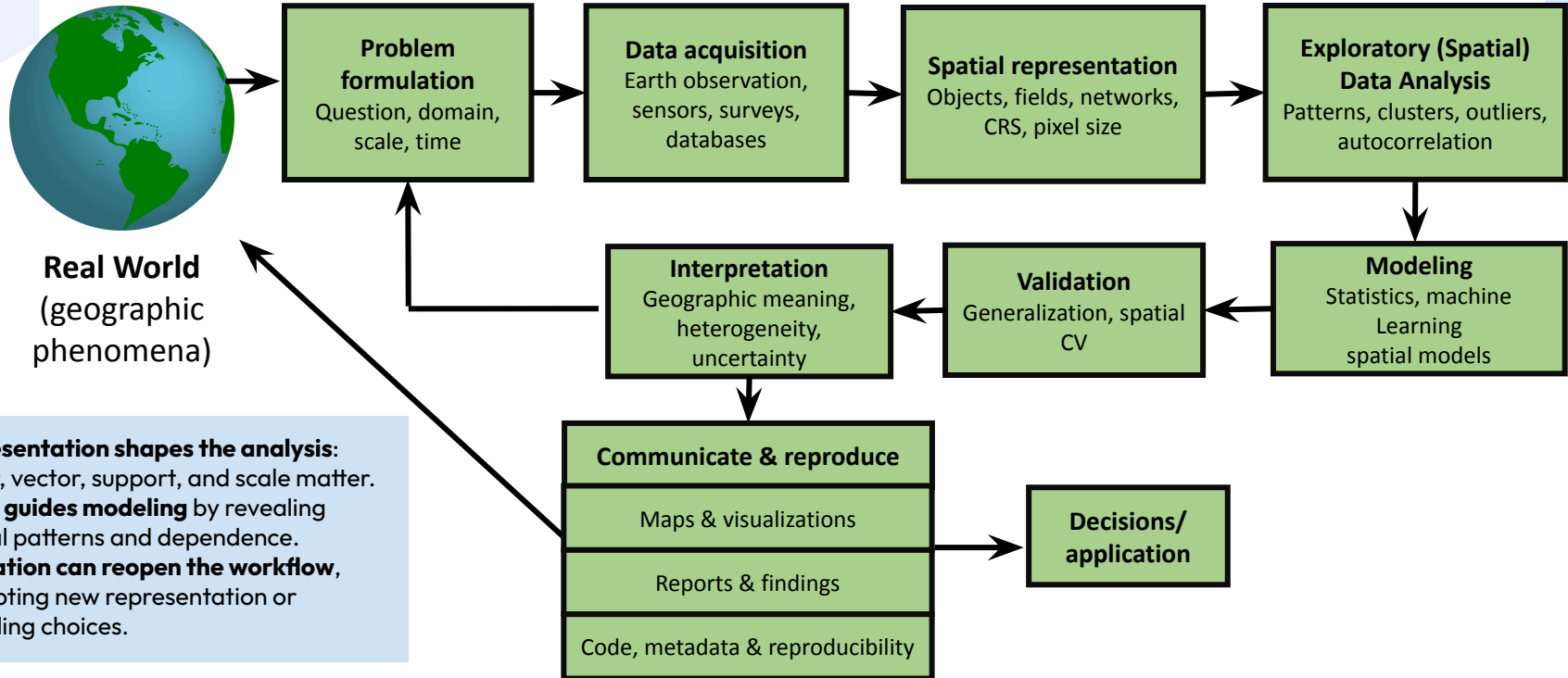


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# **The Workflow in Geospatial Data Science**

# What is Geospatial Data Science?

Geospatial Data Science is an iterative process



*Iterative, not strictly linear.*

# Spot the problem

Look at the case below and identify a potential methodological issue

## CASE

A team wants to predict deforestation across the Amazon.

- They collect **5,000 training samples** concentrated near roads.
- They randomly split **80% for training** and **20% for testing**.
- The model achieves **95% accuracy**.



## Your task

There is more than one valid answer

What would you question first?

- 1) Sampling design
- 2) Spatial dependence
- 3) Validation strategy
- 4) Prediction domain

**Which two issues would you investigate first — and why?**

# What should you take away?

1 **Data science** connects data, methods, computation, and domain questions

2 Location adds **relationships**, not only coordinates

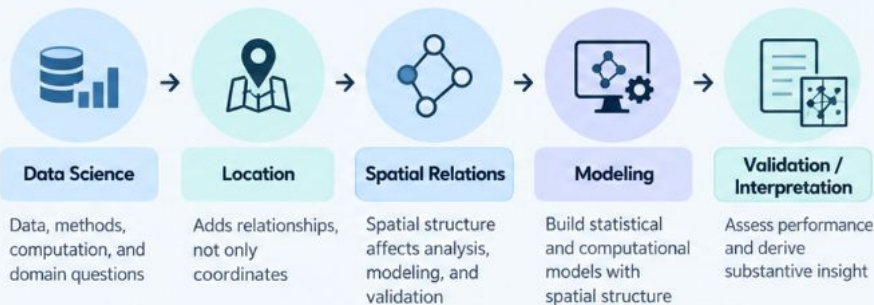
3 Spatial structure affects **analysis, modeling, and validation.**

4 **Spatial, geospatial, and geographic** perspectives emphasize different questions.

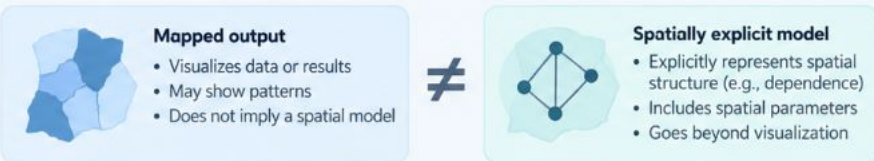
5 A mapped output is not necessarily a **spatially explicit model.**

## From Data to Insight

An integrated view of the spatial data science workflow



## A map is not a spatially explicit model

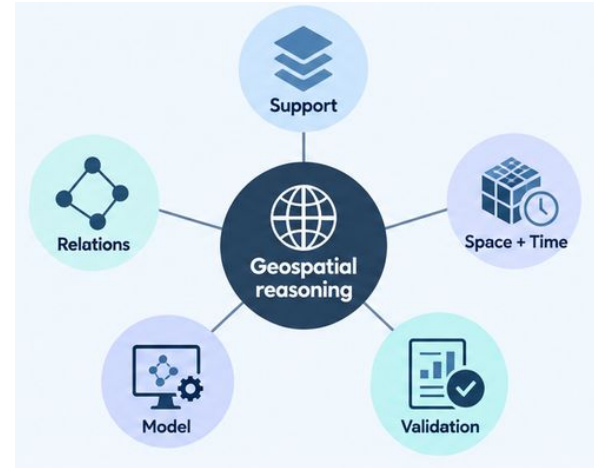


*A map can be a useful product, but it is not the same as a spatial model.*

# Think Geospatially

Before analyzing geospatial data, ask:

- 1 What is the **spatial unit** or **support**\*?
- 2 Which **spatial relationships** matter?
- 3 How are **space** and **time** represented?
- 4 Does the model actually use spatial **structure**?
- 5 Does validation match the **prediction domain**?



**Geospatial Data Science is not Data Science plus coordinates**

Space must contribute relationships, structure, and geographic meaning to the analysis.

\*Spatial support is the spatial unit or footprint over which a variable is observed, measured, or aggregated.

---

# What is Geo Data Science?

Édipo Cremon  
Rennan Marujo

